

Introduction to course

Data models in database systems

Lecture 1

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Agenda

- Passing the course
- Databases and database management systems
- Relational DBMS
- DBMS complexity
- Other approaches to data organisation

Passing the course



General information

- On-site classes + off-site work, amount roughly equal to the time spent on-site
- Course topics:
 - object-relational databases
 - multidimensional databases
 - document-oriented databases

Grading

- **50%: grade for the practical work**
 - minimum grade (4) if the assignment scores 50% and meets the minimum requirements
 - must receive **at least 10 grade points** for all the practical works (each max. 10 grade points) to get 12 access to exam
- **25%: the midterm test**
 - minimum passing grade (4) on 50% of the points.
 - midterm must be passed
- **25%: the final exam tests**
 - minimum passing grade (4) at 50% of the points.
 - final exam must be passed

Tentative deadlines

- Midterm test: November 2, final exam - January 18

Topic	Sumbmission deadline
Multidimensional databases	October 19
Object-relational databases	November 23
Document-oriented databases	December 21

Course policies

- No late submissions
- You can use 1 emergency pass for 3 days for any practical work deadline
 - You can use it for one deadline (the pass cannot be split among multiple deadlines)
 - If it is a group work, all members must agree
- No do-overs, except the last exam and once – midterm test.
- If the grade points for the practical works are not enough to pass the course, or midterm test is failed, **student has failed the course before the exam**

Course topics

- Multidimensional databases
- Object-relational databases
- Hierarchical databases

Course schedule

Weeks	Topic (schedule may fluctuate)	Deliverables
Week 1 (01.09. – 07.09.)	Data model in database systems overview.	
Week 2 (08.09. – 14.09.)	ORDBS	
Week 3 (15.09. – 21.09.)	ORDBS	
Week 4 (22.09. – 28.10.)	ORDBS	
Week 5 (29.09. – 05.10.)	ORDBS	
Week 6 (06.10. – 12.10.)	Consultation for submission 1	
Week 7 (13.10. – 19.10.)	MDDBS	Submission 1
Week 8 (20.10. – 26.10.)	MDDBS	
Week 9 (27.10. – 02.11.)	MDDBS	
Week 10 (03.11. – 09.11.)	Midterm test	
Week 11 (10.11. – 16.11.)	Consultation for submission 2	
Week 12 (17.11. – 23.11.)	HDBS	Submission 2
Week 13 (24.11. – 30.11.)	HDBS	
Week 14 (01.12. – 07.12.)	HDBS	
Week 15 (08.12. – 14.12.)	Consultation for submission 3	
Week 16 (15.12. – 21.12.)	Final lecture	Submission 3
Week 17 (02.01. – 04.01.)	-	
Week 18 (05.01. – 11.01.)	Consultation	
Week 19 (12.01. – 18.01.)	Final exam	
Week 20 (19.01. – 25.01.)	Repeated final exam	

Academic integrity policy

- Academic integrity is strictly observed in this and any other RTU courses. Depending on the heaviness of the academic integrity breach, student's task may be annulled, or in worse cases or repeated offenses, student may be given a failing grade in the course. Any work that is presented as individual, must be done alone, if similarities are found, both works will be annulled regardless of who submitted it first.
- No using internet sources without referencing, including the code (references may be put in comments). No direct copying from internet sources.

Databases and database management systems

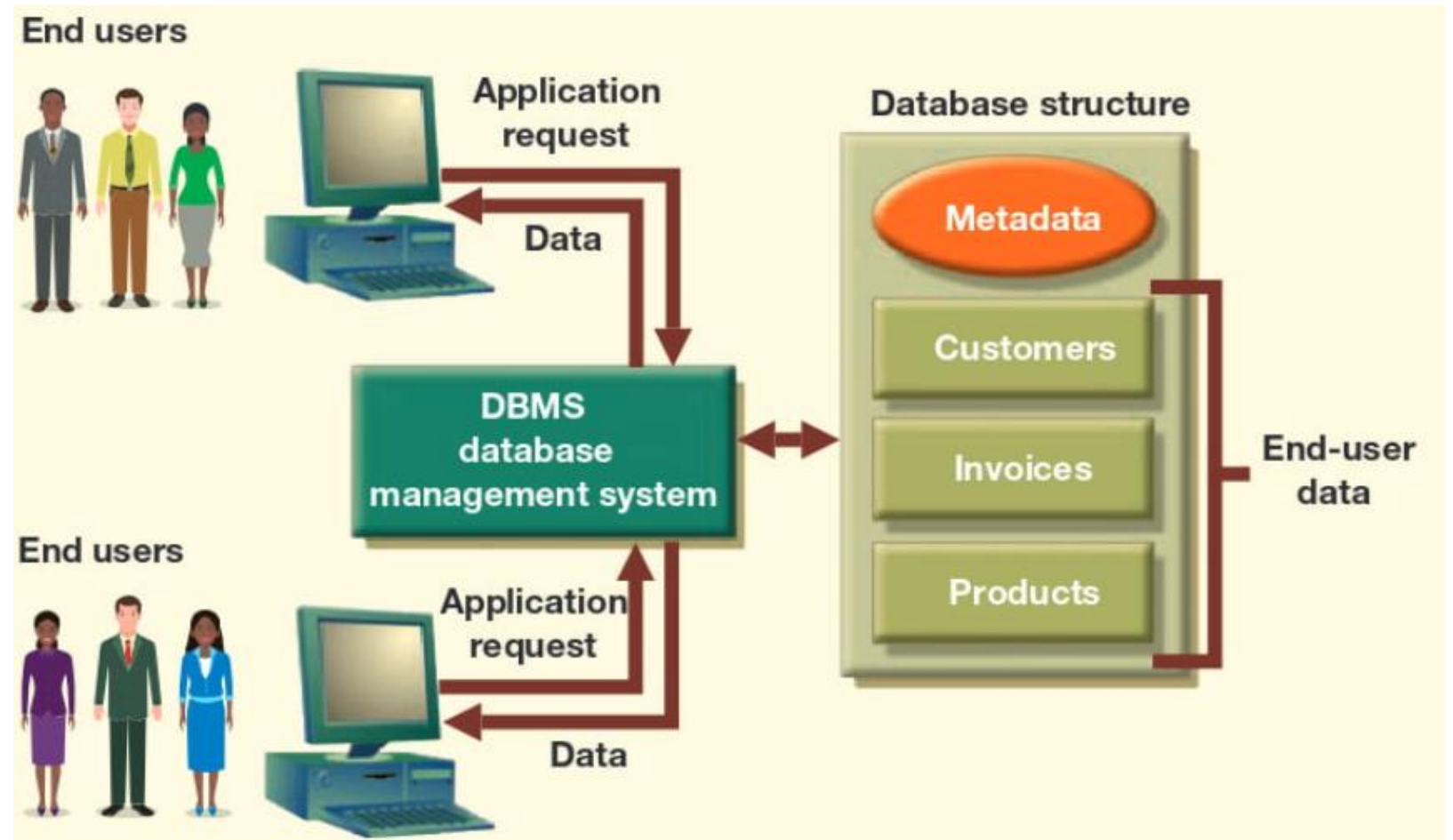


Database

- **Database** is a collection of interconnected data
- It represents some aspect of the real world
- Database is the collection of coherent data that has some meaning
- Database is designed, implemented and filled with some specific goal in mind

Database vs. Database management system

- **Database** is a collection of data
- **Database management system** consists of database software (and the database itself)



**How did data solutions
develop: file systems and
database systems**



Data base system development

- Out of necessity, clearly business driven
- Initially there were file systems – all data were stored in one large file
 - In different stages these systems evolved – from one file to file structures

Could have looked like this

Order no.	Client no.	Name	Surname	Address	Code	Item	Colour	Qt.	Price
672	8	John	Smith	3 Maple Blw, London	86175	Box	white	1	10
672	8	John	Smith	3 Maple Blw, London	76291	Notebook	white	2	10
673	16	Ana	Doe	15 Maple Blw, London	86175	Box	white	3	9
673	16	Ana	Doe	15 Maple Blw, London	76291	Notebook	white	2	9
673	16	Ana	Doe	15 Maple Blw, London	76293	Notebook	red	1	9

Drawbacks – technology-wise

- Did not guarantee that the data will not be lost
- No simultaneous data access for multiple users
- Complex system administration
- Structural dependence – updating records changed the structure
- Data dependence – e.g., changing data format corrupted the system
- Data redundancy
- **Technologically the main drawback was the long search for a particular record or data object**

Drawbacks – organization-wise

- Long development
- Difficulty in quick answers
- Extensive programming even for minute tasks

**Database systems – a
solution!**



Relational databases



The concept of the relation

- Relation demonstrates the association between two attributes, such as

Student (student, course)

- In the **schema level** we know a student will have a course
- In the **instance level** we know which student has which course

Structure of the table

	Name				
	Value				

Field

Record

Example of the table

ST_NAME	ST_SURNAME	ST_PERS_ID	ST_STUD_ID	ST_GROUP
John	Smith	010188-00000	000RDB001	1
Peter	Walsh	010189-00001	000RDB002	2
William	Green	010277-00009	001RDB002	2
Anna	Ivanova	050566-00000	000RDB003	4

Primary key

- The **primary key** is a column that uniquely identifies a record
- Practically, each table in the DB needs such a field (or set of fields)
- The type of the primary key column can be scalar, usually a number
- Primary keys are of critical importance to create relationships among tables

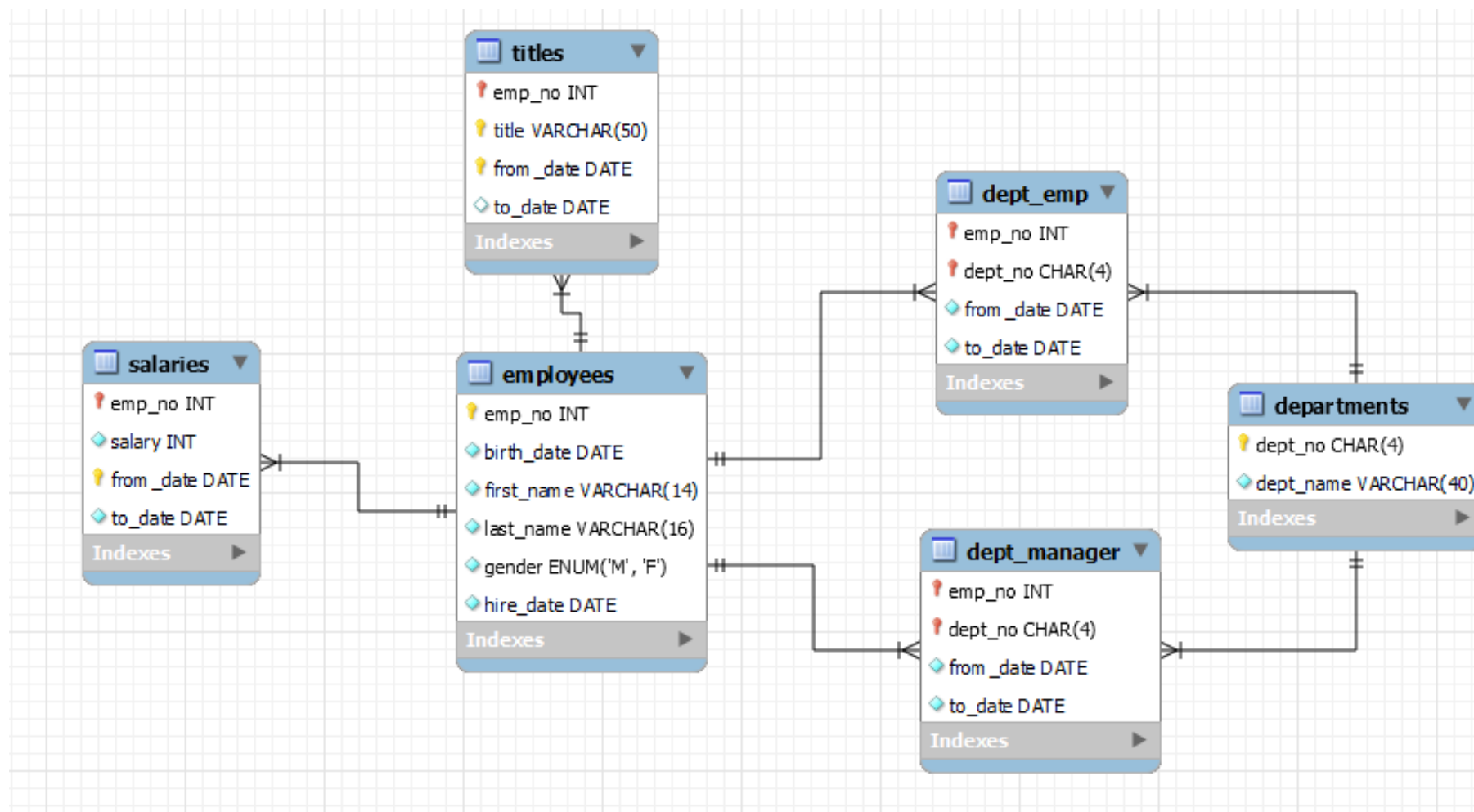
Foreign key

- **Foreign key** enables connecting tables and referring to another table
- The referred attribute must be a primary key

Design and manipulation of data in relation model



ER diagrams



SQL (Structured Query Language)

- SQL (Structured Query Language) is a language used for defining and manipulating data in databases
- Query language SQL was developed in IBM at the beginning of eighties.
- Currently, it includes more options, such as view definition, transaction control, authorisation etc.

A simple **SELECT** query

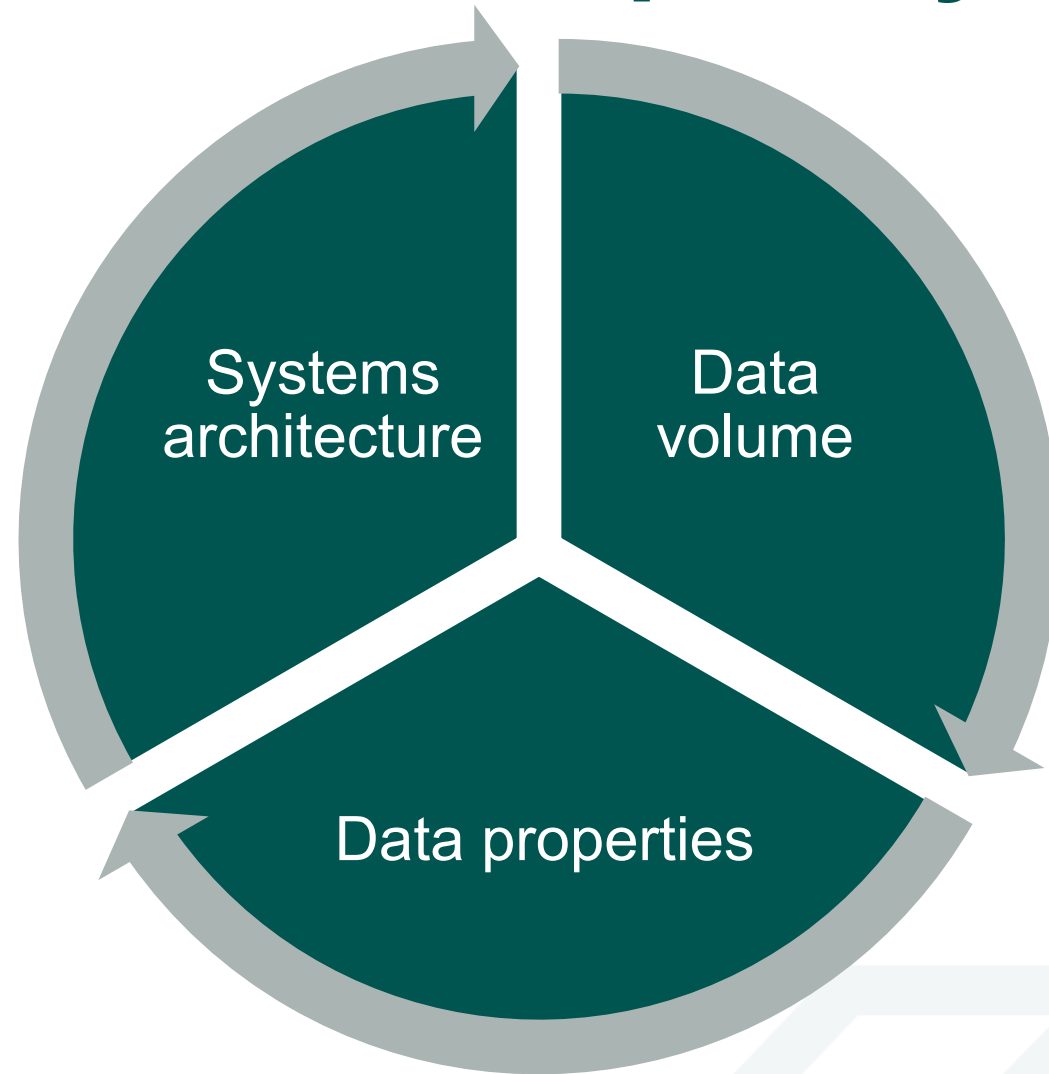
```
SELECT *  
FROM Workers;
```

```
SELECT Name, Surname  
FROM Workers  
WHERE Dep_ID = 3;
```

The complexity of the database systems

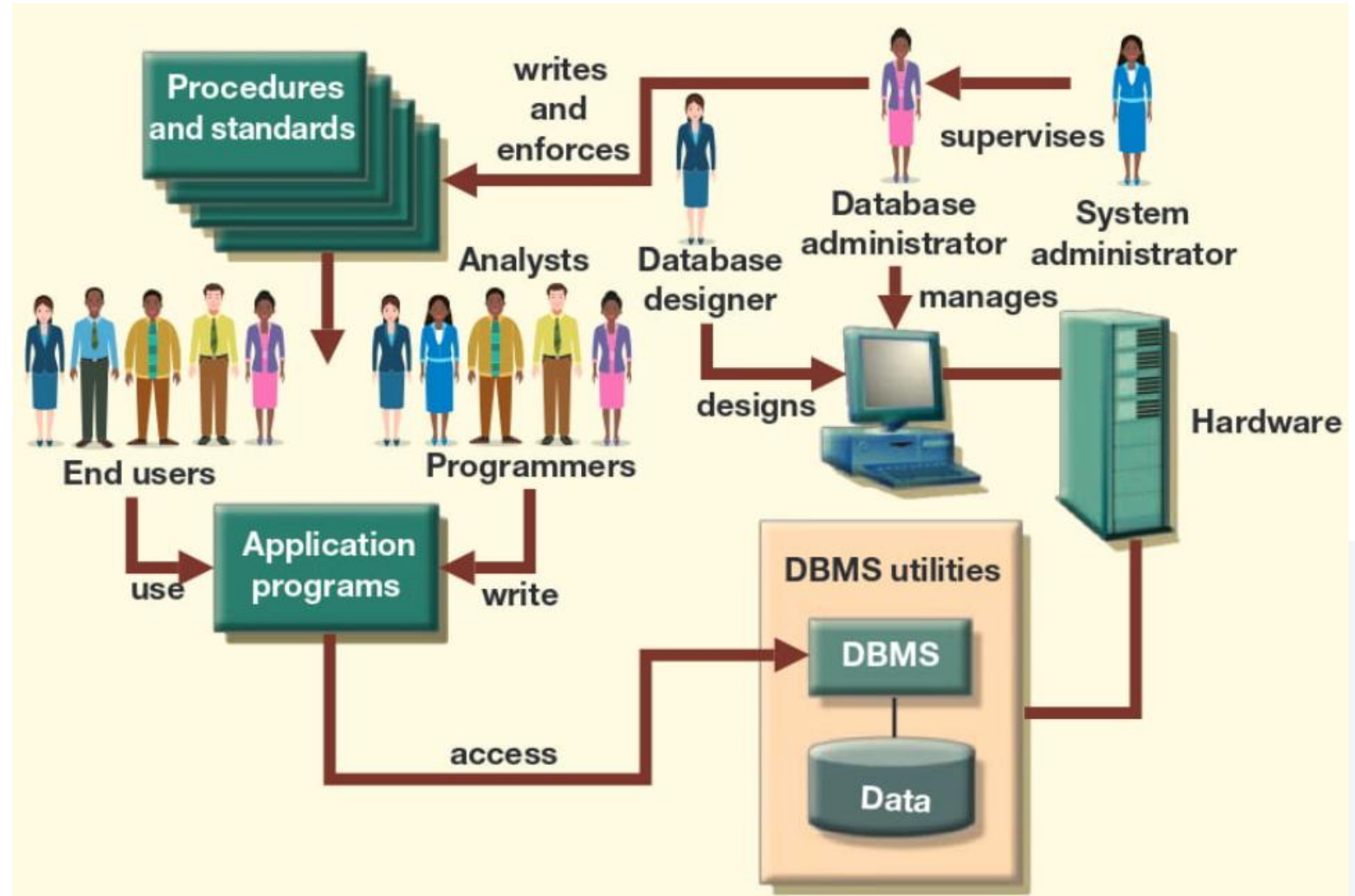


What adds to the complexity?



Database system environment

- The term **database system** refers to an organisation of components that define and regulate the collection, storage, management and use of data within a database environment.



Managing complexity

- **Abstraction** means that there are several separated levels on how we view the database:
- Physical level – hardware; how the data is stored
- Logical level – logical structure, **schema** of the database
- View level – describes only part of the database

The volume of the data



1 kilobyte	1,000 000,000,000,000,000
1 megabyte	1,000,000 000,000,000,000,000
1 gigabyte	1,000,000,000 000,000,000,000
1 terabyte	1,000,000,000,000 000,000,000
1 petabyte	1,000,000,000,000,000 000,000
1 exabyte	1,000,000,000,000,000,000 000
1 zettabyte	1,000,000,000,000,000,000,000

<https://www.statista.com/statistics/871513/worldwide-data-created/#:~:text=The%20total%20amount%20of%20data,replicated%20reached%20a%20new%20high.>

How large are the databases about which we are talking?

- Is it the size/storage?
 - Small: 10^5 or fewer records.
 - Medium: 10^5 to 10^7 records.
 - Large: 10^7 to 10^9 records.
 - Very large: 10^9 or greater number of records.
-
- What are the problems of such databases?

How does the size add to challenges?

- More users
- More storage space needed
- More suited logical schemas needed
- **Not only the size but also data properties have changed!**

Data properties and complexity

- **Integrated** – minimal redundancy (data duplication)
- **Shared** – database can be used by multiple users – even simultaneously
- **Valuable** – business value
- Relatively **large**

...new:

- **Complex**
- **Variable** structure



**Relational model still is the
best for simple use cases**

Drawbacks of relational model

- Scalable vertically, not horizontally
- Only supports atomic data types
- The schema is somewhat fixed

Other logical schemas



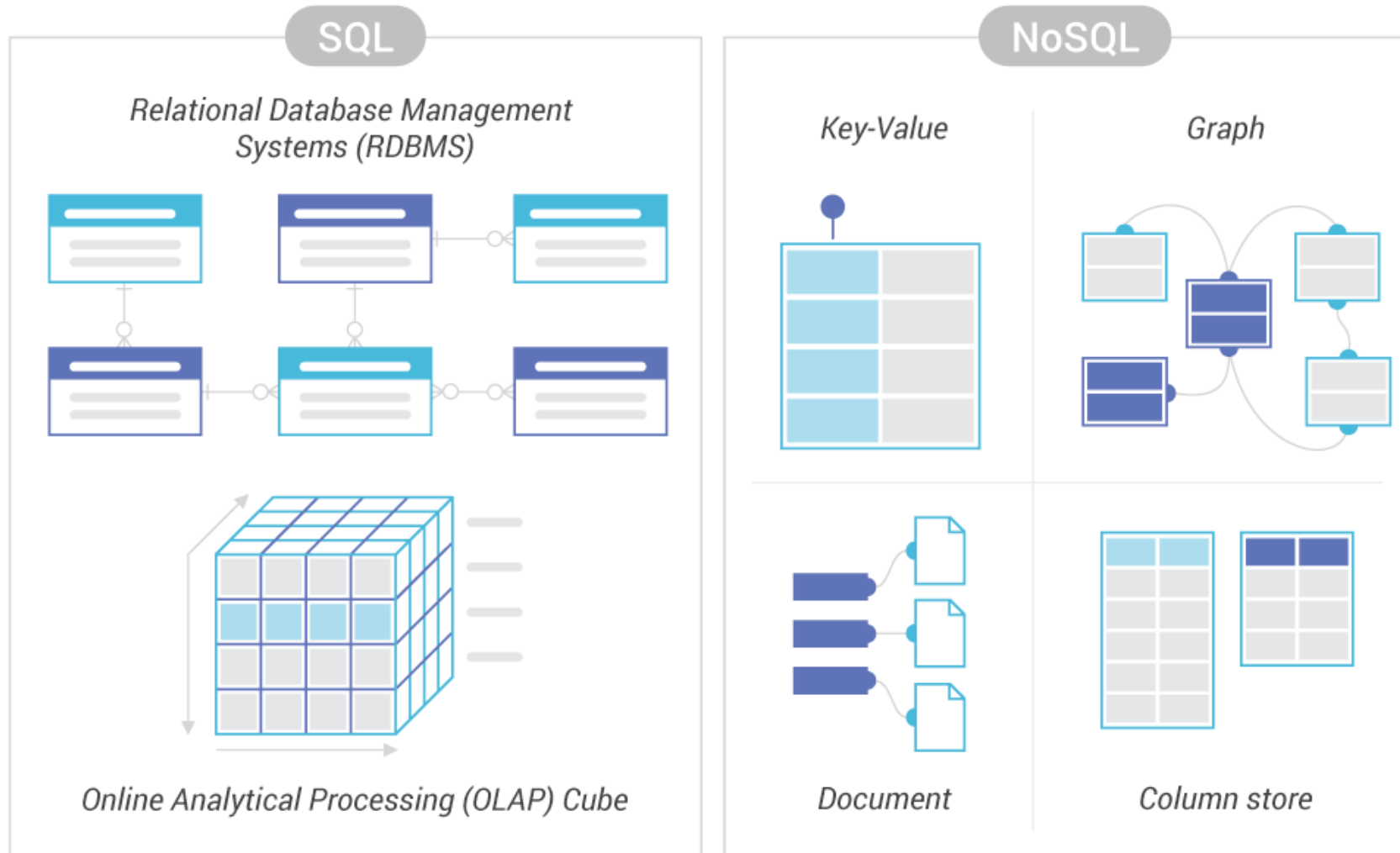
Buzzwords and content behind them

- NoSQL
- Big data
- ...and many more

Alternatives/additions to relational model – are they all dead?

- Hierarchical data model
 - Network data model
 - Object data model
-
- NoSQL databases?

Options for data logical organisation



Referenced books

[1] <https://www.amazon.com/Database-Principles/dp/1473768047>

[2] <https://db-book.com/>